

- 26 Here is a prism $ABCDEFGH$ with a horizontal, rectangular base $ABGF$

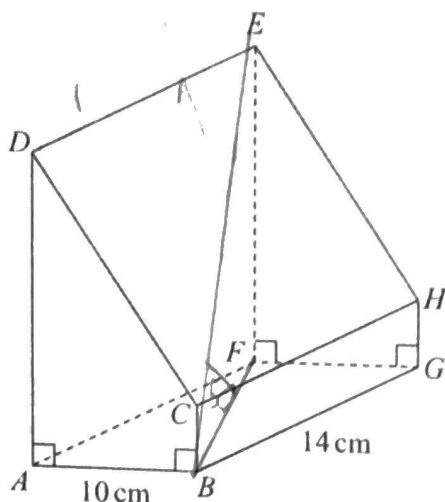


Diagram NOT
accurately drawn

$$AB = 10 \text{ cm} \quad BG = 14 \text{ cm}$$

$$\text{angle } DAB = \text{angle } ABC = \text{angle } EFG = \text{angle } FGH = 90^\circ$$

$$BC = \frac{1}{5} AD$$

$$\text{Volume of Prism} = \text{C.S Area} \times l$$

The angle of elevation of E from B is 50°

Calculate the volume of the prism.

Give your answer correct to 3 significant figures.

Show your working clearly.

$$AB = FB = 10 \text{ (Opposite sides of rectangle)}$$

$$\text{since } ABGF \rightarrow \square \quad FGB = 90^\circ$$

$\therefore$ Pythagoras

$$FB^2 = BG^2 + FG^2$$

$$FB = \sqrt{14^2 + 10^2} = 2\sqrt{74}$$

$$\therefore \tan 50 = \frac{EF}{BF}$$

$$\tan 50 = \frac{EF}{2\sqrt{74}}$$

$$EF = 2\sqrt{74} \times \tan 50$$

$$EF = 20.503$$

$$\therefore EF = DA = 20.503$$

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA



$$\therefore PC = \frac{1}{5} \times 20.503$$

$$= 4.1006$$

C.S Area

$$\text{Area of trapezium} = \frac{h(a+b)}{2}$$

$$= \frac{10(20.503 + 4.1006)}{2}$$

$$= 123.018$$

$$\therefore \text{Volume} = 123.018 \times 14$$

$$= 1,722.252$$

$$= \underline{\underline{1720}}$$

1,720 cm³

(Total for Question 26 is 5 marks)

Turn over for Question 27